

KSHITIZ KOTHARI

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Full-Stack Engineer specializing in low-latency distributed systems, real-time data synchronization, algorithmic optimization. Hands-on expertise with MERN stack, C++, Web Assembly, and ML-integrated backends.

SKILLS

Languages	C++, JavaScript
CSE Fundamentals	Data Structures and Algorithm, OOP, OS, RDBMS, CN
Web Technologies	JS, React.js, Node.js, Express.js
Databases	Oracle SQL, MongoDB, MySQL
Familiar Technologies & Tools	Git, GitHub, Postman, Figma, Wix Studio

WORK EXPERIENCE

Freelance Software Engineer / Self-Employed | [GoRafts](#)

Remote | May 2024 – Sep 2025

- Architected and delivered a production-grade, responsive adventure booking platform using **React (Vite)**, Tailwind CSS, and Framer Motion, enabling the client to onboard **5+ new tour routes** and penetrate localized travel markets.
- Achieved a **40% reduction in page load time** by implementing code-splitting, lazy-loading strategies, and an automated image asset compression pipeline, directly improving Core Web Vitals scores.
- Eliminated manual booking overhead by integrating an **EmailJS-powered inquiry automation pipeline**, reducing checkout friction and achieving **100% on-time client delivery** across all project milestones.
- Implemented end-to-end responsive design with cross-browser compatibility, ensuring consistent UX across mobile, tablet, and desktop form factors.

PROJECTS

COSKETCH [Visit GitHub](#)

(React.js, Node.js, Socket.io, C++, Web Assembly, MongoDB)

- Built a real-time collaborative whiteboard featuring a hybrid architecture of a **C++ Web Assembly (Wasm) CRDT engine**, Node.js, and Socket.IO, enabling fluid, lag-free concurrent drawing for multiple active users.
- Optimized canvas hit-testing from **O(N)** to **O(log N)** by designing a custom **2D Quadtree spatial index**, preventing rendering drops and ensuring **60 FPS** performance even with thousands of concurrent elements.
- Eliminated data-race conditions and state divergence by implementing a custom **C++ CRDT** with fractional indexing compiled to **Wasm**, guaranteeing absolute deterministic multi-user synchronization without central database locking.

KODA [Visit GitHub](#)

(React.js, Node.js, Express, Flask, Python, Scikit-Learn, MongoDB)

- Architected a **microservices Web IDE (KODA)** using React 19, Node.js/Express, Flask, and MongoDB, establishing a secure sandbox environment for live code evaluation and performance analysis.
- Developed a **C++ runtime execution tracer** in a Flask sandbox to capture variable mutations, streaming instant visual execution logs **via Socket.IO Web Sockets** to reduce debugging friction.
- Engineered an **ML static analysis pipeline** using **Scikit-Learn (Random Forest)** to parse AST features, providing developers with immediate, accurate time and space complexity classification before code execution.

EDUCATION

Graphic Era Hill University
Master of Computer Applications (MCA)

Expected August 2027
CGPA : 8.14/10

Graphic Era Hill University
Bachelor of Computer Applications (BCA)

July 2022 – August 2025
CGPA : 8.15/10

ACHIEVEMENTS

- Competed in HackWithInfy 2026**, successfully advancing to Round 2 of Infosys' national premier coding championship among top engineering talents.
- Mastered cloud and foundational AI concepts** to earn 6 professional credentials during the [Microsoft AI Skills Fest](#), specializing in Azure solution deployment, Microsoft Foundry models, Computer Vision, and Natural Language Processing.
- Executed comprehensive training to secure the Coursera Professional Certification in [Full Stack Web Development](#) by Angela Yu, establishing a formal mastery of end-to-end MERN stack architectures.